

Title	Research on Customer Churn Prediction and Model Interpretability Analysis
Harvard reference	Peng, K., Peng, Y. and Li, W. (2023) 'Research on customer churn prediction and model interpretability analysis', <i>PLOS ONE</i> , 18(12), e0289724.
Link	Research on customer churn prediction and model interpretability analysis PLOS One
Modelling	The study compared six machine-learning classification models: XGBoost, LightGBM, Decision Tree, K-Nearest Neighbours (KNN), Gradient Boosting Decision Tree (GBDT), and Extra Trees. XGBoost was then optimised using a Genetic Algorithm (GA), producing a GA-XGBoost model. The researchers also used SHAP to interpret the model's predictions.
Preprocessing techniques	The researchers performed data cleaning and transformation, removed a highly correlated feature (Avg_Utilization_Ratio), converted categorical variables using one-hot encoding, and applied Z-score normalisation. They also investigated class imbalance using oversampling and hybrid resampling approaches, including SMOTE, ADASYN and SMOTEENN.
Datasets used	The study used the Credit Card Customer dataset (BankChurners) obtained from Kaggle. The dataset contains demographic, behavioural and transaction information about bank customers and includes an attrition indicator
Key takeaways	1. Class imbalance can substantially affect customer-churn prediction, making techniques such as SMOTEENN useful to investigate. 2. XGBoost can provide strong predictive performance for customer-churn classification, particularly when combined with hyperparameter optimisation. 3. SHAP can make complex models more interpretable by showing which customer characteristics

Title	Research on Customer Churn Prediction and Model Interpretability Analysis
Harvard reference	Keramati, A., Ghaneei, H. and Mirmohammadi, S.M. (2016) 'Developing a prediction model for customer churn from electronic banking services using data mining', Financial Innovation, 2, 10.
Link	Developing a prediction model for customer churn from electronic banking services using data mining Financial Innovation Springer Nature Link
Modelling	The study used a Decision Tree to predict customer churn from electronic banking services. The authors selected the decision-tree approach because it could handle numerical and categorical variables and, importantly, produce understandable if then rules describing the characteristics of churned customers.
Preprocessing techniques	The researchers detected and removed outliers, handled missing values using mean replacement and k-nearest-neighbour imputation (k=5), addressed class imbalance using bootstrap sampling, divided the data into 70% training and 30% testing sets, and applied forward selection and backward elimination for feature selection. Backward elimination was ultimately selected and resulted in the removal of the career variable.
Datasets used	The study used a private dataset obtained from a banks database. It contained 4383 electronic-banking customers covering the period from March 2013 to March 2015. Variables included demographic characteristics, electronic-banking transaction activity, customer relationship length and customer complaints. A customer with no electronic banking transactions for at least two years was classified as a churner.
Key takeaways	1. Customer transaction activity and engagement can be useful predictors of future customer disengagement. 2. Data cleaning, missing-value treatment and class-imbalance handling should be considered before modelling. 3. Interpretable models such as decision trees can help identify characteristics associated with customer dormancy/churn and can therefore support customer-retention strategies.

Title	Prediction of Bank Credit Customers Churn based on Machine Learning and Interpretability Analysis
Harvard reference	Li, Y. and Yan, K. (2025) 'Prediction of bank credit customers churn based on machine learning and interpretability analysis', <i>Data Science in Finance and Economics</i> , 5(1), pp. 19–34.
Link	Prediction of bank credit customers churn based on machine learning and interpretability analysis
Modelling	The study evaluated six machine-learning methods: Random Forest (RF), Gradient Boosting Decision Tree (GBDT), Extra Trees, AdaBoost, XGBoost and CatBoost. Four sampling approaches were investigated: Random Oversampling, SMOTE, Borderline-SMOTE and ADASYN. SHAP was used for model interpretation, while an R-learner was used for causal analysis
Preprocessing techniques	The researchers removed the customer identifier and irrelevant Naive Bayes output columns, converted categorical variables into numerical representations, replaced “Unknown” categories with the majority class for selected categorical variables, normalised independent variables, and divided the dataset into an 80% training and 20% test set. The training data was balanced using four sampling techniques.
Datasets used	The study used a bank customer churn dataset downloaded from Kaggle, containing 10127 customer records. The dataset includes demographic information, credit-card characteristics and customer transaction/behavioural information.
Key takeaways	1. Class imbalance is an important consideration in customer-churn modelling, and different resampling techniques can be compared before selecting an approach. 2. XGBoost performed strongly across the different sampling approaches used in the study. 3. SHAP can be used to identify important behavioural variables, including transaction counts, transaction amounts and the number of products held by customers.

Publicly Available Dataset Description

Credit Card Customers / BankChurners Dataset

Name	Credit Card Customers (BankChurners)
Harvard reference	Goyal, S. (n.d.) <i>Credit Card Customers</i> . Kaggle. Available at: https://www.kaggle.com/datasets/sakshigoyal7/credit-card-customers
Link	Credit Card customers
Number of instances	10127 customer records. The dataset does not provide a predefined train/test split. The researcher will create the training and testing partitions during Part B.
Number of feature columns	The original file contains 23 columns in total. This includes the customer identifier, target variable and two Naive Bayes classifier output columns. For modelling, the identifier and the two Naive Bayes output columns are not treated as predictive features. The remaining variables contain demographic, financial, account and behavioural information.
Number of classes (if applicable)	2 classes: Existing Customer and Attrited Customer. There are approximately 8,500 existing customers and 1,627 attrited customers.
Target range (if applicable)	Not applicable because this is a classification problem. The target variable is Attrition_Flag, indicating whether the customer is an existing or attrited customer.
Origin	The dataset is a publicly available bank customer dataset hosted on Kaggle. It contains demographic, financial, credit-card and behavioural information about customers. Variables include age, gender, education, income category, card category, months on book, number of products, inactive months, transaction counts, transaction amounts, credit limit and credit utilisation. The dataset was made available to support analysis of customer attrition and prediction of customers likely to leave.